Predicting Tomorrow's Wildfire: Tree Models and Neural Networks Compared on Real Fire Data

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Abstract—Knowing where a wildfire will burn tomorrow helps crews and evacuation planners. Using the Next Day Wildfire Spread (NDWS) dataset, we predict whether each 1 km square near a fire will be burning the next day. We compare tree models that use 32 hand-designed features, two neural networks (a compact CNN and a U-Net), and hybrids that combine the two. We test the models on three increasingly difficult splits of the data: new weeks, new places, and both together. The best tree model scored highest on all three tests, with a PR-AUC of 0.265, 0.141 and 0.124. The best hybrid selected without using test data scored 0.259, 0.140 and 0.115: it improved on the U-Net on the two harder tests but did not beat the trees. We also show that a model can inflate its score by learning that the middle of each image usually burns, and we prevent this. Today's fire map was by far the most important input. Our main conclusion is that well-designed features remain strong on this problem, and that the way a model is tested changes what can be concluded.

Index Terms—wildfire spread, gradient boosting, U-Net, remote sensing, data leakage, hybrid models

I. Introduction

Wildfire teams need to know where a fire may be tomorrow so they can place crews and plan evacuations. Traditional simulators such as FARSITE [1], based on Rothermel's fire spread model [2], need detailed information about fuel and wind and are slow to run many times. Machine learning offers another route: learn from past fires using satellite and weather data.

This project asks a simple question: can neural networks that see a whole map find patterns that tree models using hand-designed features miss? We answer it on the NDWS dataset [3], and we take care to test the models fairly.

Our main findings are:

- Tree models scored highest on all three tests. Neural networks came close but did not beat them.
- Hybrids improved on the U-Net on the two harder tests but did not beat the best tree model.
- About 95% of the standard test areas also appear in the training data, so harder tests are needed.
- A model can gain a large but misleading score from learning that the centre of each image tends to burn.

II. Problem Statement

For each 1 km square in a 64×64 km area around a fire, we estimate the chance that the square is burning the next day. We use only information known by the end of today. We call this the active target. We also report a transition target, which looks only at squares not burning today, so it measures new fire alone. Only about 1.10% of training squares are burning the next day, so the problem is very imbalanced. The models find statistical patterns; they do not explain how a fire physically spread.

III. Research Objectives

1) Build strong tree-based baselines using hand-designed features.
2) Build neural networks that use the whole map.
3) Check whether results hold on new places and new weeks.
4) Measure how much a model can gain from patch position alone.
5) Test whether a hybrid improves on both approaches, and report the result either way.

IV. Literature Review

Early fire models were based on physical equations [2], [1]. Jain et al. [4] survey how machine learning is now used in wildfire science, and early deep learning studies [5], [6] showed that CNNs can learn spread from map data. Sayad et al. [7] predicted fire occurrence from satellite features.

Huot et al. [3] released NDWS, reporting PR-AUC values of 11.5% for persistence, 19.8% for logistic regression, 22.5% for random forest and 28.4% for a convolutional autoencoder. WildfireSpreadTS [8] extends this task to several days.

Gradient boosting [9], as in XGBoost [10] and LightGBM [11], and random forests [12] remain strong on tabular data, and Grinsztajn et al. [13] found that trees often beat neural networks on such data. For images, fully convolutional networks [14] and the U-Net [15] give an output for every pixel, and dilated convolutions [16] let a network see further without shrinking the image. Dice loss [17] is a common choice when the target is small.

Kayhan and van Gemert [18] and Liu et al. [19] show that convolutional networks can learn where they are in an image, which matters for our position test. Stacking [20] combines models using data they did not train on.

For evaluation, Saito and Rehmsmeier [21] show why precision and recall are more informative than ROC curves when positives are rare, and Roberts et al. [22] warn that random splits can overstate performance on map data.

V. Dataset Description

NDWS contains 18,545 image patches of 64×64 squares at 1 km resolution, from the contiguous US between 2012 and 2020 [3]. Each patch has 11 inputs from today: elevation, wind direction and speed, minimum and maximum temperature, humidity, rainfall [23], drought index, vegetation (NDVI), population density and an energy release index. It also has today's fire map. The label is the next day's fire map.

Each square is marked as fire, no fire or unknown (for example, because of cloud). About 2.3% of labels are unknown, and we ignore those squares. The public files have no coordinates, dates or fire names. The published split gives 14,979 training, 1,877 validation and 1,689 test patches.

VI. Data Preprocessing

Both model families use the same cleaned data:

- Impossible values, such as temperatures outside 200–330 K, are marked as missing.
- Rainfall is kept at zero or above, vegetation is rescaled, and population is log-transformed.
- For neural networks, each input is scaled using training data only. Wind and slope direction are stored as east and north parts, so that flipping an image also flips those values correctly.
- The public files do not say which way is north. We found it from the data: only with one orientation is new fire clearly more common downwind than upwind (ratio 1.12, 95% range 1.09 to 1.15).

VII. Exploratory Data Analysis

About 74% of next-day fire squares were not burning today, so simply repeating today's fire map finds only about 26% of them. Fig. 1 shows that new fire is much more likely close to existing fire.

VIII. Feature Engineering

A tree model sees one square at a time, so we give each square 32 features that describe its surroundings (Table I). Every feature uses only today's data. For example, wind alignment measures whether the wind is blowing from the nearest fire toward the square. Vapour-pressure deficit, a measure of how dry the air is, follows the FAO formula [24].

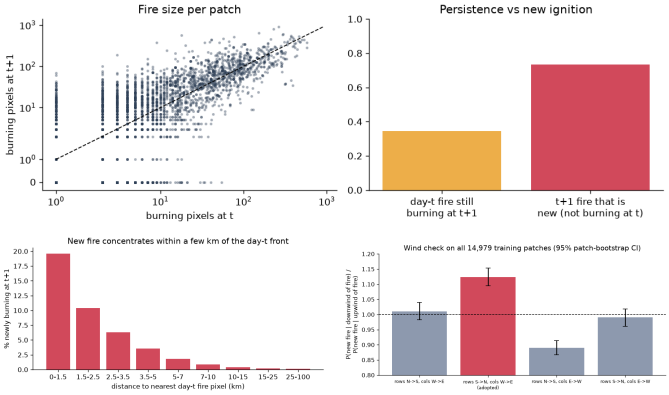


Fig. 1. How fire behaves in the data: how fire stays or moves (top), chance of fire by distance to today's fire (left), and the wind test used to find north (right).

TABLE I
The 32 tabular features

Group	Count	Features
Raw	10	elevation, wind speed, temperature min/max, humidity, rainfall, drought, NDVI, log population, ERC
Fire	8	today's fire and unknown flags, share of fire in windows of 3, 5, 9 and 17, distance to nearest fire, log fire count
Wind	6	east/north wind, fire up/downwind within 3 km, fire–wind alignment, wind push
Terrain	4	slope, aspect (sin, cos), upslope from fire
Weather	2	temperature range, vapour-pressure deficit
Vegetation	2	5×5 NDVI mean and standard deviation

IX. AML Methodology

We use simple baselines: Dummy (always predicts the average fire rate), persistence (fire stays where it is) and neighbourhood persistence. We then train logistic regression, random forest, XGBoost and LightGBM.

Non-fire squares vastly outnumber fire squares, so training keeps every fire square and 2% of the others. This makes the raw scores too high, so we correct them. With sampling rate r and raw score p_s, the corrected probability [25] is

$$p = \frac{r\,p_s}{r\,p_s + 1 - p_s}. \tag{1}$$

Boosted trees stop adding trees when validation performance stops improving, and we try a few tree sizes on validation data.

X. Deep Learning Architecture

Both networks produce a prediction for every square. The compact CNN uses three blocks whose filters look further and further out, so each prediction sees a 29-square-wide area. It has 81,097 parameters. The U-Net first shrinks the map to capture the big picture, then expands it back to full size. Skip connections carry fine detail straight across, so the edges of a fire stay sharp.

TABLE II
The three tests

Test	New weeks	New places	Train / val / test
Official	Yes	No	14,979 / 1,877 / 1,689
Spatial	No	Yes	14,822 / 1,855 / 1,868
Strict	Yes	Yes	12,050 / 389 / 405

It has 1,086,841 parameters. Batch normalisation [26] and dropout [27] help training and reduce overfitting.

We train with weighted cross-entropy, which makes missing a fire more costly. With true label y, predicted chance p and weight w, the loss is

$$\ell = -\left[w\, y \log p + (1-y)\log(1-p)\right]. \tag{2}$$

Training uses AdamW [28], stops early when validation performance stops improving, and flips and rotates images. It also uses random 32×32 crops, which hide the position of each patch's centre (Section XII-B).

XI. Hybrid Methodology

Hybrids try to combine the strengths of both approaches, because trees have useful hand-made features and networks can see shapes. We test two kinds:

- Feature-assisted U-Net: the U-Net receives the hand-made features as extra inputs, for 31 input channels and 1,090,729 parameters.
- Late fusion: a small logistic regression learns how much to trust a tree and a network:

$$\operatorname{logit} p = a + b_1 \operatorname{logit} p_{\text{tree}} + b_2 \operatorname{logit} p_{\text{net}}. \tag{3}$$

Here logit is the log of the odds of fire. Following stacking [20], the weights are fitted on half of the validation data, which the base models did not train on. The other half chooses the best hybrid. Test data is never used for this choice.

XII. Experimental Setup

A. Three fair tests

Elevation barely changes between days, so two patches with identical terrain are the same place. Using this, we grouped patches into 845 locations. We found that about 95% of the standard test patches share terrain with the training data. Table II shows our three tests. A separate tuning split chooses the network settings, so no test patch influences them.

B. Blocking the position shortcut

Every patch is centred on a fire. A model could learn that "the middle burns", which would not help in a real forecast [18]. We use three protections: networks train on random crops; every model is also scored on fixed random windows where the centre is not special; and we measure the effect directly with a position-only predictor and a U-Net trained on full images.

TABLE III
Test PR-AUC at Phase 1 and in the final re-run

Model	Official (P1)	Official (final)	Spatial (P1)	Spatial (final)	Strict (new)
Persistence	0.107	0.107	0.037	0.037	0.027
Logistic regression	0.230	0.230	0.117	0.117	0.095
Random forest	0.265	0.265	0.135	0.135	0.120
XGBoost	0.265	0.265	0.141	0.141	0.115
LightGBM	0.263	0.263	0.138	0.138	0.124

TABLE IV
Terms used in this report

Term	Meaning
Patch	One 64×64 km map of a fire
PR-AUC	How well a model finds real fires without too many false alarms; higher is better
Precision	When the model says fire, how often it is right
Recall	Of the real fires, how many the model finds
Lift	PR-AUC divided by the fire rate; how much better than random guessing
Leakage	When a model uses information it would not have in real use
95% interval	A range showing how uncertain a score is, computed by resampling test patches 500 times
Paired test	Compares two models on the same resampled patches
Ablation	Removing one part to see how much it matters

C. Tuning and repeats

We tune one network setting at a time on the tuning split: model type, loss, dropout, weight decay, width and input channels. The final networks are trained three times with different random seeds. Hybrids use the first seed. Everything runs from one script, scripts/run_all.sh, with settings in configuration files.

D. Changes since Phase 1

The graded Phase 1 materials remain unchanged on the phase1-final-submission branch. A full re-run reproduced the official and spatial scores to three decimals (Table III). We made three additions: a strict split, because the Phase 1 spatial split still mixed weeks; a position-leak test, because Phase 1 did not test whether tabular models use position; and a Dummy baseline, CNN, tuned U-Net and hybrids.

XIII. Evaluation Metrics

Our main score is PR-AUC, computed on every known square. Random guessing scores about the fire rate, which differs between tests, so scores should be compared within a test. We also report recall at fixed false-alarm rates, confusion matrices, and other metrics in the Appendix. Table IV explains the terms used in this report.

XIV. Results

A. Main comparison

Table V and Fig. 2 give the main results. The best tree model scored highest on all three tests. Network rows use

TABLE V
Test PR-AUC with 95% intervals. Higher is better.

Model	New weeks (official)	New places (spatial)	New weeks and places (strict)
Dummy (training rate)	0.013 [0.012, 0.014]	0.005 [0.005, 0.005]	0.006 [0.005, 0.007]
Persistence	0.107 [0.097, 0.118]	0.037 [0.032, 0.041]	0.027 [0.021, 0.035]
Logistic regression	0.230 [0.213, 0.249]	0.117 [0.106, 0.127]	0.095 [0.079, 0.113]
Random forest	0.265 [0.248, 0.282]	0.135 [0.123, 0.149]	0.120 [0.097, 0.145]
XGBoost	0.265 [0.248, 0.283]	0.141 [0.128, 0.154]	0.115 [0.094, 0.141]
LightGBM	0.263 [0.246, 0.281]	0.138 [0.127, 0.150]	0.124 [0.103, 0.148]
Compact CNN	0.256 [0.240, 0.272]	0.129 [0.117, 0.141]	0.093 [0.071, 0.119]
U-Net	0.257 [0.242, 0.272]	0.130 [0.118, 0.142]	0.101 [0.079, 0.124]
U-Net + engineered channels	0.254 [0.239, 0.270]	0.129 [0.117, 0.141]	0.116 [0.091, 0.141]
Hybrid (selected on validation)	0.259 [0.244, 0.276]	0.140 [0.127, 0.152]	0.115 [0.093, 0.140]

the first training seed; variation between seeds is covered below.

B. Are the differences real?

The intervals in Table V overlap, so we use paired tests against XGBoost, the tree model chosen on validation data for all three tests. The U-Net is lower by -0.008, -0.011 and -0.014, and all three differences are statistically significant. So the networks' gap is small but real. LightGBM has the highest strict test score, but the paired test uses XGBoost because it was chosen on validation data. Full results are in Appendix Tables VII–X.

C. Scoring without the position shortcut

On fixed random windows, position gives almost no information: a position-only predictor has a lift of just 1.1, 1.1 and 1.1. Here the gap narrows. On the official test the hybrid beats a window-trained LightGBM by +0.012, which is the only significant hybrid win over a tree. On the spatial test the models are level. On the strict test the tree stays ahead of the U-Net by -0.015 and of the hybrid by -0.009. Details are in Tables XI and XII.

D. New fire only

For squares not burning today (Table XIII), the U-Net beats XGBoost by +0.005 on the official test, but trails it by -0.012 and -0.012 on the spatial and strict tests. The network's small lead does not carry over to places it has not seen.

E. The position shortcut

A predictor that uses only position scores 0.135, 0.121 and 0.137 on full images. On the strict test it is higher than every valid model. A U-Net trained on full images gains +0.084, +0.103 and +0.089. We report these only as warnings, not as real results.

For tree models (Table XIV), adding position as a feature helps a lot, so our features do not already contain it. A separate model can partly recover closeness to the centre from the features (ROC-AUC 0.843, 0.742 and 0.753), most likely because fire itself is near the centre. Trees trained on full images and on windows score similarly, which suggests that the tree results do not depend on position.

TABLE VI
U-Net ablations on the tuning split

Change	Val. PR-AUC	Change vs selected
Selected U-Net	0.2386	–
No flips or rotations	0.2417	+0.0032
No dropout	0.2334	-0.0051
No weight decay	0.2330	-0.0056
None of the three	0.2305	-0.0081
No skip connections	0.2278	-0.0108
No day-t fire channels	0.0297	-0.2089
No wind channels	0.2325	-0.0060
No weather channels	0.2274	-0.0111
No elevation or NDVI	0.2284	-0.0102

F. Hybrids

The chosen hybrid was Fusion: tree + feature U-Net on the official and spatial tests and Fusion: tree + U-Net on the strict test (Table XV). It was -0.006 below XGBoost on the official test, which is significant, and level with it on the other two (-0.001 and -0.001, with intervals containing zero). It beat the U-Net on the two harder tests. So the hybrid improves the network but not the trees.

G. Different training seeds

Table XVI repeats training three times. Scores vary by a standard deviation of 0.001 to 0.013, which is similar to several gaps between models. Adding features to the U-Net gave the largest gain on the strict test (mean 0.116 against 0.104), but it also varied the most, so we treat it as a hint rather than a proven gain.

H. Maps and errors

Fig. 3 shows one example. Models work well near today's fire and miss fire that moves further away. XGBoost finds 0.95, 0.72 and 0.66 of fire within 1.5 km of today's fire on the three tests, but only 0.45, 0.19 and 0.06 between 1.5 and 3.5 km, and almost none beyond that. More examples are in Fig. 5 and Fig. 6.

XV. Ablation Studies

An ablation removes one part to see how much it matters. Table VI and Fig. 4 show the U-Net results. Removing today's fire map drops the score from 0.239 to 0.030. Other removals changed it only slightly: no skip connections gave 0.228, no weather gave 0.227 and no wind gave 0.233. Removing image flips gave 0.242, slightly above the chosen model, so we do not claim that augmentation helped. Each is a single run, so small differences should not be over-read.

For tree models (Table XVII), raw values alone are weak (0.010 on spatial and 0.012 on strict), and fire features alone reach 0.111 and 0.095.

XVI. Discussion

Why did trees stay ahead? The ablations show that tomorrow's fire depends mostly on today's fire and its immediate surroundings, and our features describe exactly that. Networks can learn these patterns too, but only

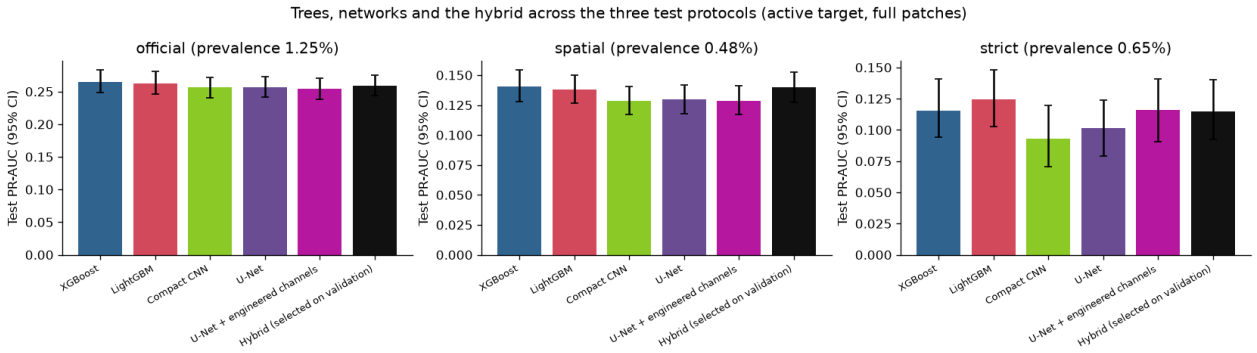


Fig. 2. Test PR-AUC with 95% intervals for the main models on each of the three tests.

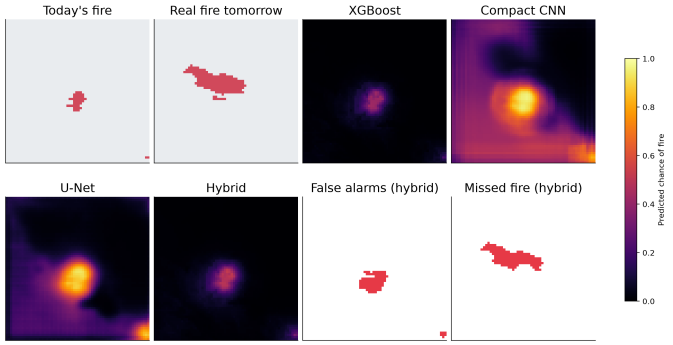


Fig. 3. One strict-test example, showing today's fire, tomorrow's real fire, each model's predicted chance of fire, and the hybrid's false alarms and missed fire.

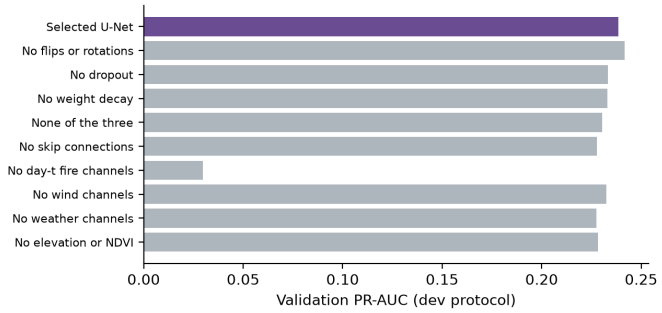


Fig. 4. Effect of each ablation on the U-Net's validation PR-AUC.

have about 10^4 patches to learn from. This agrees with Grinsztajn et al. [13], although we did not test this explanation directly.

How much does the test matter? Moving from the official test to tests on new places roughly halves PR-AUC, partly because fire is rarer in those tests. Trees stay ahead of networks in the full-image scores, so shared terrain does not explain their lead.

Position. Full-image scores include a large position effect, so scores from different studies can only be compared when they handle position in the same way. Our scores are also not directly comparable with published NDWS scores, because training and evaluation details differ.

Hybrids. Combining a tree and a network improved on the network in the two harder tests, but did not beat the best tree.

In practice, models mostly find fire next to today's fire and rarely detect spread several kilometres away.

XVII. Limitations

- NDWS has no coordinates, dates, fire names, fuel moisture or firefighting information.
- Weeks are assigned at random, so we test new weeks but not future seasons.
- Weather covers all of today, so a forecast can only be issued after the day ends.
- Satellite fire labels can be wrong because of cloud, smoke and timing.
- We can only detect exactly matching terrain, so nearby fires may appear in both training and test data.
- The strict test has only 405 patches, so its results are uncertain.
- Every sample contains a fire, so results do not apply to monitoring all land.
- Tuning and ablation results come from single runs.

XVIII. Conclusion

Tree models with well-designed features were the strongest models on all three tests. The CNN, U-Net and hybrids came close but did not beat them. The hybrid improved on the U-Net on the two harder tests and did not beat the best tree, so the hybrid idea was tested and not supported as an improvement over trees. The strict test and the position tests show that scores depend strongly on how data is split and sampled.

XIX. Future Work

Useful next steps are to use WildfireSpreadTS [8], which has real dates and fire names; to try models that use several days, such as ConvLSTM [29]; to add fuel-moisture data; and to make the predicted probabilities more reliable for operational use.

Reproducibility and Contributions

All code, settings and results are in the project repository. scripts/run_all.sh reproduces every experiment, and every number here is generated from the result files. Individual contributions are in CONTRIBUTIONS.md.

Appendix A
Detailed Results

These tables and figures give the full detail behind the results in the main text.

TABLE VII
Official test (new weeks), active target, full images. Thresholded metrics use validation thresholds.

Model	PR-AUC [95% CI]	Lift	ROC-AUC	Prec.	Rec.	F1	R@1%	R@5%
Dummy (training rate)	0.013 [0.012, 0.014]	1.0	0.500	0.013	1.000	0.025	0.010	0.050
Persistence	0.107 [0.097, 0.118]	8.5	0.633	0.357	0.273	0.309	0.276	0.305
Logistic regression	0.230 [0.213, 0.249]	18.4	0.903	0.246	0.487	0.327	0.363	0.665
Random forest	0.265 [0.248, 0.282]	21.1	0.901	0.261	0.498	0.343	0.389	0.679
XGBoost	0.265 [0.248, 0.283]	21.1	0.911	0.242	0.528	0.332	0.389	0.680
LightGBM	0.263 [0.246, 0.281]	21.0	0.905	0.245	0.525	0.334	0.387	0.680
Compact CNN	0.256 [0.240, 0.272]	20.4	0.897	0.242	0.529	0.332	0.385	0.675
U-Net	0.257 [0.242, 0.272]	20.5	0.892	0.275	0.486	0.351	0.392	0.677
U-Net + engineered channels	0.254 [0.239, 0.270]	20.3	0.915	0.258	0.500	0.340	0.380	0.677
Hybrid (selected on validation)	0.259 [0.244, 0.276]	20.7	0.917	0.254	0.514	0.340	0.385	0.682

TABLE VIII
Spatial test (new places), active target, full images.

Model	PR-AUC [95% CI]	Lift	ROC-AUC	Prec.	Rec.	F1	R@1%	R@5%
Dummy (training rate)	0.005 [0.005, 0.005]	1.0	0.500	0.005	1.000	0.010	0.010	0.050
Persistence	0.037 [0.032, 0.041]	7.6	0.572	0.224	0.146	0.177	0.152	0.186
Logistic regression	0.117 [0.106, 0.127]	24.2	0.867	0.193	0.272	0.226	0.374	0.639
Random forest	0.135 [0.123, 0.149]	28.1	0.885	0.191	0.304	0.234	0.382	0.627
XGBoost	0.141 [0.128, 0.154]	29.1	0.899	0.230	0.263	0.246	0.382	0.628
LightGBM	0.138 [0.127, 0.150]	28.6	0.896	0.229	0.261	0.244	0.382	0.621
Compact CNN	0.129 [0.117, 0.141]	26.6	0.846	0.193	0.293	0.233	0.366	0.556
U-Net	0.130 [0.118, 0.142]	26.9	0.841	0.192	0.299	0.234	0.370	0.580
U-Net + engineered channels	0.129 [0.117, 0.141]	26.6	0.873	0.196	0.277	0.230	0.345	0.571
Hybrid (selected on validation)	0.140 [0.127, 0.152]	29.0	0.893	0.220	0.273	0.244	0.380	0.607

TABLE IX
Strict test (new weeks and places), active target, full images.

Model	PR-AUC [95% CI]	Lift	ROC-AUC	Prec.	Rec.	F1	R@1%	R@5%
Dummy (training rate)	0.006 [0.005, 0.007]	1.0	0.500	0.006	1.000	0.013	0.010	0.050
Persistence	0.027 [0.021, 0.035]	4.2	0.554	0.194	0.112	0.142	0.118	0.154
Logistic regression	0.095 [0.079, 0.113]	14.8	0.868	0.182	0.210	0.195	0.279	0.560
Random forest	0.120 [0.097, 0.145]	18.6	0.856	0.207	0.190	0.198	0.311	0.545
XGBoost	0.115 [0.094, 0.141]	17.8	0.866	0.222	0.185	0.202	0.320	0.554
LightGBM	0.124 [0.103, 0.148]	19.3	0.879	0.204	0.214	0.209	0.324	0.596
Compact CNN	0.093 [0.071, 0.119]	14.4	0.831	0.183	0.244	0.209	0.270	0.446
U-Net	0.101 [0.079, 0.124]	15.7	0.783	0.200	0.198	0.199	0.296	0.496
U-Net + engineered channels	0.116 [0.091, 0.141]	17.9	0.811	0.203	0.266	0.230	0.319	0.493
Hybrid (selected on validation)	0.115 [0.093, 0.140]	17.8	0.853	0.222	0.192	0.206	0.316	0.552

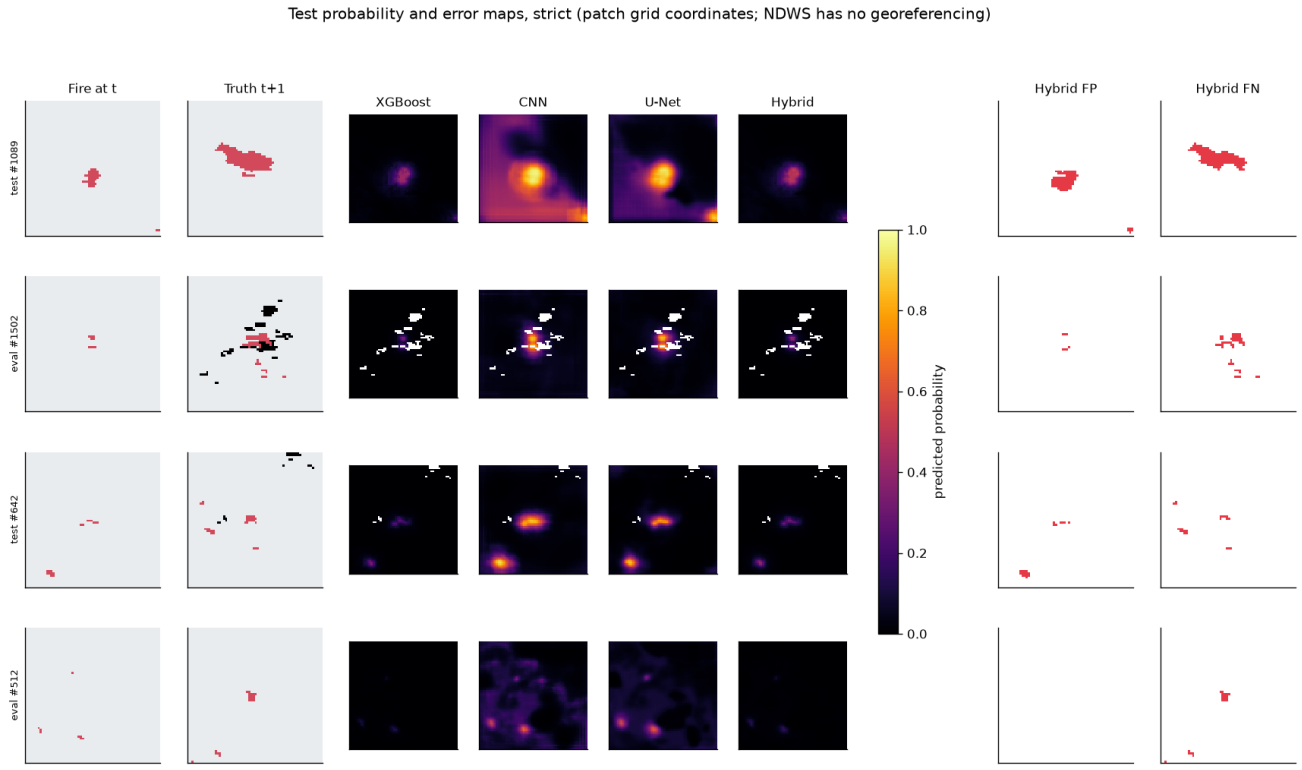


Fig. 5. More strict-test maps: today's fire, tomorrow's real fire, XGBoost, CNN, U-Net, hybrid, and hybrid false alarms and misses. Axes are grid positions because NDWS has no coordinates.

TABLE X
Paired differences in test PR-AUC, full images (first seed).

Protocol	Comparison	ΔPR-AUC	95% CI	$P(\Delta \leq 0)$
official	unet - xgboost	-0.0083	[-0.0143, -0.0023]	1.000
official	cnn - xgboost	-0.0088	[-0.0144, -0.0031]	0.998
official	unet_features - xgboost	-0.0108	[-0.0155, -0.0068]	1.000
official	hybrid - xgboost	-0.0057	[-0.0096, -0.0025]	1.000
official	hybrid - unet	+0.0027	[-0.0023, +0.0080]	0.186
official	unet - cnn	+0.0005	[-0.0040, +0.0048]	0.392
official	unet_features - unet	-0.0025	[-0.0077, +0.0025]	0.832
official	unet_fullpatch - unet	+0.0844	[+0.0760, +0.0931]	0.000
spatial	unet - xgboost	-0.0108	[-0.0152, -0.0072]	1.000
spatial	cnn - xgboost	-0.0121	[-0.0181, -0.0061]	1.000
spatial	unet_features - xgboost	-0.0120	[-0.0171, -0.0070]	1.000
spatial	hybrid - xgboost	-0.0008	[-0.0039, +0.0024]	0.724
spatial	hybrid - unet	+0.0100	[+0.0070, +0.0138]	0.000
spatial	unet - cnn	+0.0012	[-0.0027, +0.0049]	0.268
spatial	unet_features - unet	-0.0012	[-0.0052, +0.0033]	0.696
spatial	unet_fullpatch - unet	+0.1026	[+0.0937, +0.1122]	0.000
strict	unet - xgboost	-0.0141	[-0.0199, -0.0096]	1.000
strict	cnn - xgboost	-0.0222	[-0.0303, -0.0137]	1.000
strict	unet_features - xgboost	+0.0005	[-0.0056, +0.0056]	0.466
strict	hybrid - xgboost	-0.0005	[-0.0025, +0.0011]	0.738
strict	hybrid - unet	+0.0135	[+0.0104, +0.0176]	0.000
strict	unet - cnn	+0.0082	[+0.0008, +0.0142]	0.020
strict	unet_features - unet	+0.0146	[+0.0088, +0.0210]	0.000
strict	unet_fullpatch - unet	+0.0886	[+0.0724, +0.1045]	0.000

TABLE XI
Test results on fixed 32×32 windows.

Official

Model	PR-AUC [95% CI]	Lift	ROC-AUC
Position prior in window (leak probe)	0.032 [0.030, 0.036]	1.1	0.549
LightGBM (full-patch model)	0.293 [0.274, 0.316]	10.3	0.876
LightGBM (window-trained)	0.290 [0.273, 0.310]	10.1	0.871
Compact CNN	0.287 [0.269, 0.307]	10.0	0.867
U-Net	0.297 [0.278, 0.316]	10.4	0.876
U-Net + engineered channels	0.284 [0.265, 0.304]	9.9	0.853
Hybrid (selected on validation)	0.301 [0.283, 0.321]	10.5	0.878

Spatial

Model	PR-AUC [95% CI]	Lift	ROC-AUC
Position prior in window (leak probe)	0.015 [0.013, 0.016]	1.1	0.537
LightGBM (full-patch model)	0.174 [0.159, 0.189]	12.9	0.877
LightGBM (window-trained)	0.169 [0.154, 0.184]	12.5	0.853
Compact CNN	0.161 [0.145, 0.176]	11.9	0.822
U-Net	0.167 [0.150, 0.182]	12.3	0.840
U-Net + engineered channels	0.167 [0.151, 0.183]	12.3	0.832
Hybrid (selected on validation)	0.172 [0.155, 0.187]	12.7	0.855

Strict

Model	PR-AUC [95% CI]	Lift	ROC-AUC
Position prior in window (leak probe)	0.020 [0.016, 0.024]	1.1	0.535
LightGBM (full-patch model)	0.157 [0.128, 0.188]	8.7	0.847
LightGBM (window-trained)	0.148 [0.116, 0.182]	8.2	0.797
Compact CNN	0.123 [0.094, 0.157]	6.8	0.792
U-Net	0.134 [0.104, 0.166]	7.4	0.770
U-Net + engineered channels	0.143 [0.112, 0.177]	8.0	0.778
Hybrid (selected on validation)	0.139 [0.109, 0.172]	7.7	0.781

TABLE XII
Paired differences on fixed windows (first seed).

Protocol	Comparison	ΔPR-AUC	95% CI	$P(\Delta \leq 0)$
official	unet - lightgbm_crop	+0.0068	[-0.0005, +0.0142]	0.046
official	unet_features - lightgbm_crop	-0.0060	[-0.0119, -0.0006]	0.986
official	hybrid - lightgbm_crop	+0.0116	[+0.0058, +0.0171]	0.000
official	hybrid - unet	+0.0048	[+0.0026, +0.0070]	0.000
spatial	unet - lightgbm_crop	-0.0027	[-0.0070, +0.0013]	0.864
spatial	unet_features - lightgbm_crop	-0.0022	[-0.0069, +0.0028]	0.816
spatial	hybrid - lightgbm_crop	+0.0020	[-0.0008, +0.0052]	0.108
spatial	hybrid - unet	+0.0047	[+0.0013, +0.0080]	0.002
strict	unet - lightgbm_crop	-0.0145	[-0.0234, -0.0080]	1.000
strict	unet_features - lightgbm_crop	-0.0049	[-0.0128, +0.0023]	0.912
strict	hybrid - lightgbm_crop	-0.0091	[-0.0160, -0.0039]	1.000
strict	hybrid - unet	+0.0054	[+0.0038, +0.0075]	0.000

TABLE XIII
PR-AUC on the transition target (new fire only).

Model	official	spatial	strict
Dummy (training rate)	0.009	0.004	0.006
Persistence	0.009	0.004	0.006
Logistic regression	0.123	0.078	0.075
Random forest	0.135	0.087	0.087
XGBoost	0.141	0.090	0.084
LightGBM	0.138	0.089	0.092
Compact CNN	0.143	0.073	0.062
U-Net	0.145	0.078	0.072
U-Net + engineered channels	0.138	0.075	0.081
Hybrid (selected on validation)	0.142	0.087	0.084

TABLE XIV
Position-leak tests of LightGBM on test data.

Test	official	spatial	strict
Features only, PR-AUC	0.263	0.138	0.124
Features + position, PR-AUC	0.345	0.278	0.251
Recover position from features, ROC-AUC	0.843	0.742	0.753
Window: full-patch model, PR-AUC	0.293	0.174	0.157
Window: window-trained model, PR-AUC	0.290	0.169	0.148
Window: position prior, PR-AUC	0.032	0.015	0.020
Window: positive rate	0.029	0.014	0.018

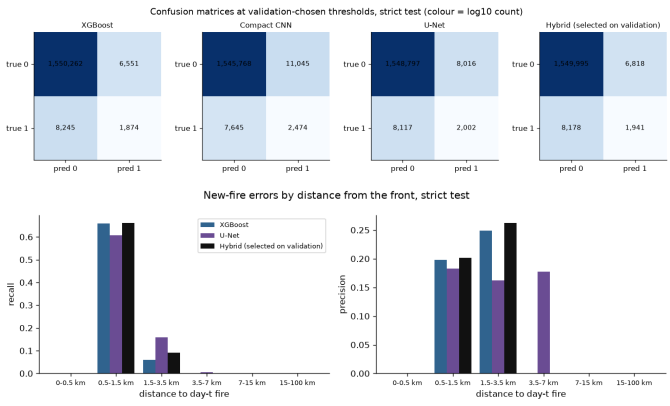


Fig. 6. Strict-test confusion matrices (top) and errors by distance to today's fire (bottom).

TABLE XV
Hybrid candidates. Weights are fitted on half of validation data, and the other half selects the hybrid.

Protocol	Candidate	Weights on logits	Val. PR-AUC	Test PR-AUC
official	Fusion input: tree only	tree=0.91	0.192	0.265
official	Fusion input: U-Net only	unet=0.99	0.199	0.257
official	Fusion: tree + U-Net	tree=0.53, unet=0.43	0.205	0.270
official	Fusion: tree + CNN	tree=0.80, cnn=0.13	0.195	0.266
official	Fusion: tree + feature U-Net (selected)	tree=0.21, unet_features=0.81	0.205	0.259
official	U-Net + engineered channels	–	0.204	0.254
spatial	Fusion input: tree only	tree=1.02	0.228	0.141
spatial	Fusion input: U-Net only	unet=1.16	0.229	0.130
spatial	Fusion: tree + U-Net	tree=0.65, unet=0.42	0.236	0.142
spatial	Fusion: tree + CNN	tree=0.66, cnn=0.43	0.238	0.142
spatial	Fusion: tree + feature U-Net (selected)	tree=0.37, unet_features=0.85	0.242	0.140
spatial	U-Net + engineered channels	–	0.238	0.129
strict	Fusion input: tree only	tree=0.91	0.235	0.115
strict	Fusion input: U-Net only	unet=1.05	0.239	0.101
strict	Fusion: tree + U-Net (selected)	tree=0.68, unet=0.28	0.242	0.115
strict	Fusion: tree + CNN	tree=0.46, cnn=0.61	0.239	0.111
strict	Fusion: tree + feature U-Net	tree=0.42, unet_features=0.64	0.240	0.120
strict	U-Net + engineered channels	–	0.230	0.116

TABLE XVI
Test PR-AUC across three training seeds.

Protocol	Model	Mean	SD	Min	Max
official	Compact CNN	0.255	0.006	0.248	0.260
official	U-Net	0.252	0.005	0.248	0.257
official	U-Net + engineered channels	0.254	0.002	0.251	0.256
spatial	Compact CNN	0.127	0.002	0.125	0.129
spatial	U-Net	0.128	0.001	0.127	0.130
spatial	U-Net + engineered channels	0.135	0.007	0.129	0.142
strict	Compact CNN	0.099	0.007	0.093	0.107
strict	U-Net	0.104	0.003	0.101	0.108
strict	U-Net + engineered channels	0.116	0.013	0.103	0.128

TABLE XVII
LightGBM feature-group ablations, PR-AUC.

Feature groups	Spatial val.	Spatial test	Strict val.	Strict test
Raw values only	0.025	0.010	0.038	0.012
Fire context only	0.191	0.111	0.186	0.095
Raw + fire	0.212	0.130	0.214	0.116
Raw + fire + wind	0.226	0.140	0.230	0.119
Raw + fire + wind + terrain	0.228	0.142	0.224	0.119
All features	0.230	0.138	0.232	0.124
All minus raw	0.226	0.138	0.234	0.116
All minus fire	0.213	0.131	0.216	0.110
All minus wind	0.220	0.139	0.217	0.124
All minus terrain	0.227	0.139	0.234	0.119
All minus weather	0.227	0.141	0.230	0.121
All minus vegetation	0.227	0.146	0.229	0.119

TABLE XVIII
Hyperparameter sweep on the tuning split, validation PR-AUC.

Stage	Option	Val. PR-AUC	Best epoch	Best
architecture	unet	0.2356	13	✓
architecture	cnn	0.2329	17	
loss	bce	0.2356	13	
loss	weighted_bce (weight 3)	0.2268	14	
loss	weighted_bce (weight 10)	0.2386	6	✓
loss	dice	0.1728	5	
loss	bce_dice	0.2320	9	
dropout	0.0	0.2334	13	
dropout	0.2	0.2386	6	✓
dropout	0.4	0.2354	17	
weight_decay	0.0	0.2330	8	
weight_decay	0.0001	0.2386	6	✓
weight_decay	0.001	0.2281	6	
width	16	0.2339	8	
width	24	0.2386	6	✓
width	32	0.2279	13	
feature_channels	base	0.2386	6	
feature_channels	fire	0.2391	7	
feature_channels	fire_wind	0.2414	3	
feature_channels	all	0.2422	7	✓

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